

Tanish Dalal

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Computer Engineering student with experience in full-stack development, cloud systems, and applied machine learning.

EDUCATION

University of Michigan

B.S.E. in Computer Engineering | **GPA:** 4.00/4.00

Ann Arbor, MI

Aug. 2024 - May 2028

Honors: Dean's List, University Honors | **Societies:** Tau Beta Pi Engineering Honor Society

Relevant Coursework: Machine Learning, Data Structures & Algorithms, Computer Architecture, Linear Algebra, Embedded Systems, Discrete Mathematics, Logic Design

EXPERIENCE

Software Engineering Intern

Candela Technologies

Jul. 2026 – Aug. 2026

Ferndale, WA (Remote)

- Built multithreaded Python pipelines using LANforge REST APIs to automate wireless network validation across 15 access points, reducing data collection and test turnaround from 3 days to 4 hours.
- Synchronized a mobile test robot's position with wireless metrics (RSSI, latency) using ptp4l/phc2sys time synchronization, enabling network performance to be mapped to physical locations.
- Debugged and resolved ROS 2 communication failures between the mobile robot and LANforge test infrastructure across multi-subnet networks by tuning CycloneDDS discovery and analyzing Wireshark packet captures.
- Integrated 2D LiDAR and IMU sensors into the robot's ROS 2 Cartographer localization pipeline and automated configuration deployment across 6 test environments with Ansible.

Software Engineering Intern

ADNOC Distribution, EVCI & H2 Department

May 2025 – Jul. 2025

Abu Dhabi, UAE

- Extended an OCPP 1.6 telemetry ingestion pipeline in Python/FastAPI, writing SQL against TimescaleDB to normalize streaming `MeterValues` and `StatusNotification` events from 400+ EV chargers at 7k+ points/min.
- Built automated regression tests with `pytest` for OCPP message handling, covering malformed payloads, charger reconnections, duplicate events, and database write failures.
- Trained an XGBoost predictive-maintenance model on rolling-window features derived from that telemetry, achieving 0.78 PR-AUC and detecting hardware failures a median of 30 min before shutdown.

PROJECTS

Algorithm Practice Platform

JavaScript, React, Python, FastAPI, Docker, Kubernetes, AWS

Oct. 2025 – Present

- Built a real-time multiplayer algorithm battle platform using FastAPI and WebSockets, storing user accounts and ELO ratings in SQLite, supporting 50+ concurrent matches with <150 ms median latency for 100+ students.
- Built the React SPA for live matches, rendering streamed benchmark results in real time and visualizing side-by-side player runtime curves with Chart.js.
- Built an AI mock interviewer and problem generator using the OpenAI API, validating generated coding problems in a Docker sandbox and caching responses to reduce redundant API calls.
- Deployed the platform on AWS behind Nginx, executing untrusted workloads as Kubernetes Jobs with least-privilege RBAC and network isolation; automated testing, builds, and releases through GitHub Actions.

Computer Vision Lead & Co-Founder — Michigan Robomasters

Python, C++, C, ROS 2, OpenCV, STM32, CAN

Nov. 2025 – Present

- Lead a 6-member Agile team developing a ROS 2 auto-aiming system; trained YOLO26n on 8k+ labeled armor-plate frames to 0.58 mAP@50:95 and integrated `solvePnP` for 3D pose estimation at 45 FPS on a Jetson Orin Nano.
- Implemented a constant-velocity Kalman filter for armor-plate tracking and projectile lead compensation, reducing angular tracking error by 28% over raw per-frame detections.
- Developed autonomous navigation for the sentry robot in ROS 2, integrating FAST-LIO2 SLAM, Livox MID-360 LiDAR, and Nav2; achieved a 90% collision-free route-completion rate across dynamic multi-waypoint tests in Gazebo and on the physical robot.
- Implemented embedded gimbal control in C on an STM32, driving motors over CAN to hold aim during chassis rotation.

Robotic Arm Controls Developer — Michigan Mars Rover

C++, Python, ROS 2, RViz, NumPy

Aug. 2024 – Present

- Built a Python workspace point-cloud visualizer sampling arm reachability at 5 cm resolution with dexterity scoring and draggable RViz obstacles to rule out unreachable task configurations before physical tests.
- Engineered a closed-loop Cartesian IK controller for a 5-DOF robotic arm using proportional velocity control, shortest-path end-effector tracking, and joint-limit checks.
- Developed geometric IK solvers for a 30 Hz ROS 2 velocity pipeline, achieving <1 cm end-effector position error.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C++, C, SQL

AI/ML: OpenAI API, PyTorch, YOLO, OpenCV, XGBoost, scikit-learn, NumPy, pandas

Web & Databases: React, HTML/CSS, WebSockets, FastAPI, PostgreSQL, TimescaleDB, SQLite

Cloud & Tools: AWS, Docker, Kubernetes, Git, GitHub Actions, Linux, Nginx, pytest